

Nama: Muhammad Akmal Dwiansyah Putra

Kelas / Absen: TI_1G / 20

NIM: 254107020110

Case Method 2

Pembeli20.java:

```
package CM2;
```

```
public class Pembeli20 {  
    String nama;  
    String nohp;  
  
    Pembeli20(String nama, String hp){  
        this.nama = nama;  
        nohp = hp;  
    }  
}
```

Pesanan20.java:

```
package CM2;
```

```
public class Pesanan20 {  
    int kodePesanan;  
    String namaPesanan;  
    int harga;  
  
    Pesanan20(int kode, String nama, int harga){  
        kodePesanan = kode;  
        namaPesanan = nama;  
        this.harga = harga;  
    }  
}
```

NodePesanan20.java:

```
package CM2;
```

```
public class NodePesanan20 {  
    NodePesanan20 next;  
    NodePesanan20 prev;  
  
    Pesanan20 data;  
  
    NodePesanan20(Pesanan20 data){  
        this.data = data;  
    }  
}
```

NodePembeli20.java:

```
package CM2;
```

```
public class NodePembeli20 {  
    NodePembeli20 next;  
    NodePembeli20 prev;  
  
    Pembeli20 data;  
  
    NodePembeli20(Pembeli20 data){  
        this.data = data;  
    }  
}
```

DLLAntrian20.java:

```
package CM2;
```

```

public class DLLAntrian20 {
    NodePembeli20 headPembeli;
    NodePembeli20 tailPembeli;

    NodePesanan20 headPesanan;
    NodePesanan20 tailPesanan;

    int counter;

    DLLAntrian20(){
        headPembeli = null;
        tailPembeli = null;

        headPesanan = null;
        tailPesanan = null;

        counter = 0;
    }

    boolean isEmpty(){
        return headPembeli == null;
    }

    void addLast(Pembeli20 data){
        NodePembeli20 newNode = new NodePembeli20(data);
        counter++;
        newNode.data.noAntrian = counter;
        if (isEmpty()) {
            headPembeli = tailPembeli = newNode;
        }else{
            tailPembeli.next = newNode;
            newNode.prev = tailPembeli;
        }
    }
}

```

```

        tailPembeli = newNode;
    }
}

void print(){
    if (isEmpty()) {
        System.out.println("Linked list masih kosong");
        return;
    }

    System.out.println("=====");
    System.out.println("Daftar Antrian Pembeli");
    System.out.println("=====");
    System.out.println("No Antrian \t Nama \t\t No HP");

    NodePembeli20 current = headPembeli;
    while (current != null) {
        System.out.println();
        current.data.tampil();
        current = current.next;
    }
}

void removeFirst(Pesanan20 data){
    if (isEmpty()) {
        System.out.println("Linked list masih kosong");
        return;
    }

    NodePesanan20 newNode = new NodePesanan20(data);
    NodePembeli20 namaPembeli = headPembeli;
    if (headPembeli == tailPembeli) {
        headPembeli = tailPembeli = null;
    }
}

```

```

    }else{
        headPembeli = headPembeli.next;
        headPembeli.prev = null;
    }

    if (headPesanan == null) {
        headPesanan = tailPesanan = newNode;
    }else{
        tailPesanan.next = newNode;
        newNode.prev = tailPesanan;
        tailPesanan = newNode;
    }

    System.out.println(namaPembeli.data.nama + " telah memesan " + newNode.data.namaPesanan);
}

void printPesanan(){
    if (headPesanan == null) {
        System.out.println("Linked list masih kosong");
        return;
    }

    boolean swapped;
    do {
        swapped = false;
        NodePesanan20 current = headPesanan;
        while (current.next != null) {
            int comparison =
current.data.namaPesanan.compareToIgnoreCase(current.next.data.namaPesanan);

            if (comparison > 0) {
                Pesanan20 temp = current.data;
                current.data = current.next.data;
                current.next.data = temp;
                swapped = true;
            }
        }
    } while (swapped);
}

```

```

        }
        current = current.next;
    }
} while (swapped);

System.out.println("=====");
System.out.println("LAPORAN PESANAN (URUT NAMA PESANAN)");
System.out.println("=====");
System.out.printf("%-14s %-20s %s%n", "Kode Pesanan", "Nama Pesanan", "Harga");

NodePesanan20 current = headPesanan;

while (current != null) {
    current.data.tampil();
    current = current.next;
}
}
}

```

DLLMain20.java:

```

package CM2;

import java.util.Scanner;

public class DLLMain20 {
    static Pembeli20 Pembeli(Scanner sc){
        System.out.print("Nama      : ");
        String nama = sc.nextLine();
        System.out.print("Nomor Handphone: ");
        String noHp = sc.nextLine();

        return new Pembeli20(nama, noHp);
    }
}

```

```
}
```

```
static Pesanan20 Pesanan(Scanner sc){  
    System.out.print("Kode pesanan: ");  
    int kode = sc.nextInt();  
    sc.nextLine();  
    System.out.print("Nama pesanan: ");  
    String nama = sc.nextLine();  
    System.out.print("Harga: ");  
    int harga = sc.nextInt();  
    sc.nextLine();  
  
    return new Pesanan20(kode, nama, harga);  
}
```

```
public static void main(String[] args) {  
    Scanner sc = new Scanner(System.in);  
    DLLAntrian20 antrian = new DLLAntrian20();  
    int pilih;  
  
    Pembeli20 pembeli1 = new Pembeli20("Ainra", "08224500000");  
    Pembeli20 pembeli2 = new Pembeli20("Danra", "08224511111");  
    Pembeli20 pembeli3 = new Pembeli20("Sanri", "08224522222");  
  
    antrian.addLast(pembeli1);  
    antrian.addLast(pembeli2);  
    antrian.addLast(pembeli3);  
  
    do {  
        System.out.println("\n=== MENU DOUBLE LINKED LIST ===");  
        System.out.println("1. Tambah data");  
        System.out.println("2. Cetak antrian");
```

```
System.out.println("3. Hapus antrian dan pesan");
System.out.println("4. Laporan Pesanan");
System.out.println("0. Keluar");
System.out.print("Pilih menu: ");

pilih = sc.nextInt();
sc.nextLine();

switch (pilih) {
    case 1:
        Pembeli20 pembeli = Pembeli(sc);
        antrian.addLast(pembeli);
        break;
    case 2:
        antrian.print();
        break;
    case 3:
        Pesanan20 pesanan = Pesanan(sc);
        antrian.removeFirst(pesanan);
        break;
    case 4:
        antrian.printPesanan();
        break;
    default:
        break;
}

} while (pilih != 0);
sc.close();
}
}
```